success — Native-IMU reconstruction: method and interpretation

Separate experiment: existing mocap identification and outputs are unchanged.

Identification uses native gyro / specific force with fitted thrust, gyro and accel first-order lags.

Orientation: gyro prediction + acceleration update recur; magnetic factors affect output only.

Only initial orientation / position and initial velocity use mocap. Later mocap is read-only evaluation.

Comparison uses original mocap timestamps inside IMU support; reconstructed attitudes are Slerped.

Native timestamps and firmware first-or-changed magnetic freshness proxy are retained.

Static sensor covariance = trace(unbiased static covariance)/3 times identity; no bias subtraction.

Offline local polynomial derivatives and their overlap covariance are included in the lag fit.

Uncertainty: Bingham-derived right-angle covariance and deterministic local Gaussian propagation.

Shared acceleration/attitude information uses declared conservative covariance bounds.

These algebraic bounds do not guarantee coverage of systematic acceleration or magnetic bias.

Magnetic output can be worse on these data; no mocap-based gating, tuning or recursive feedback.

Model reconstruction adds the native-IMU residual wrench time series to the physical dynamics.

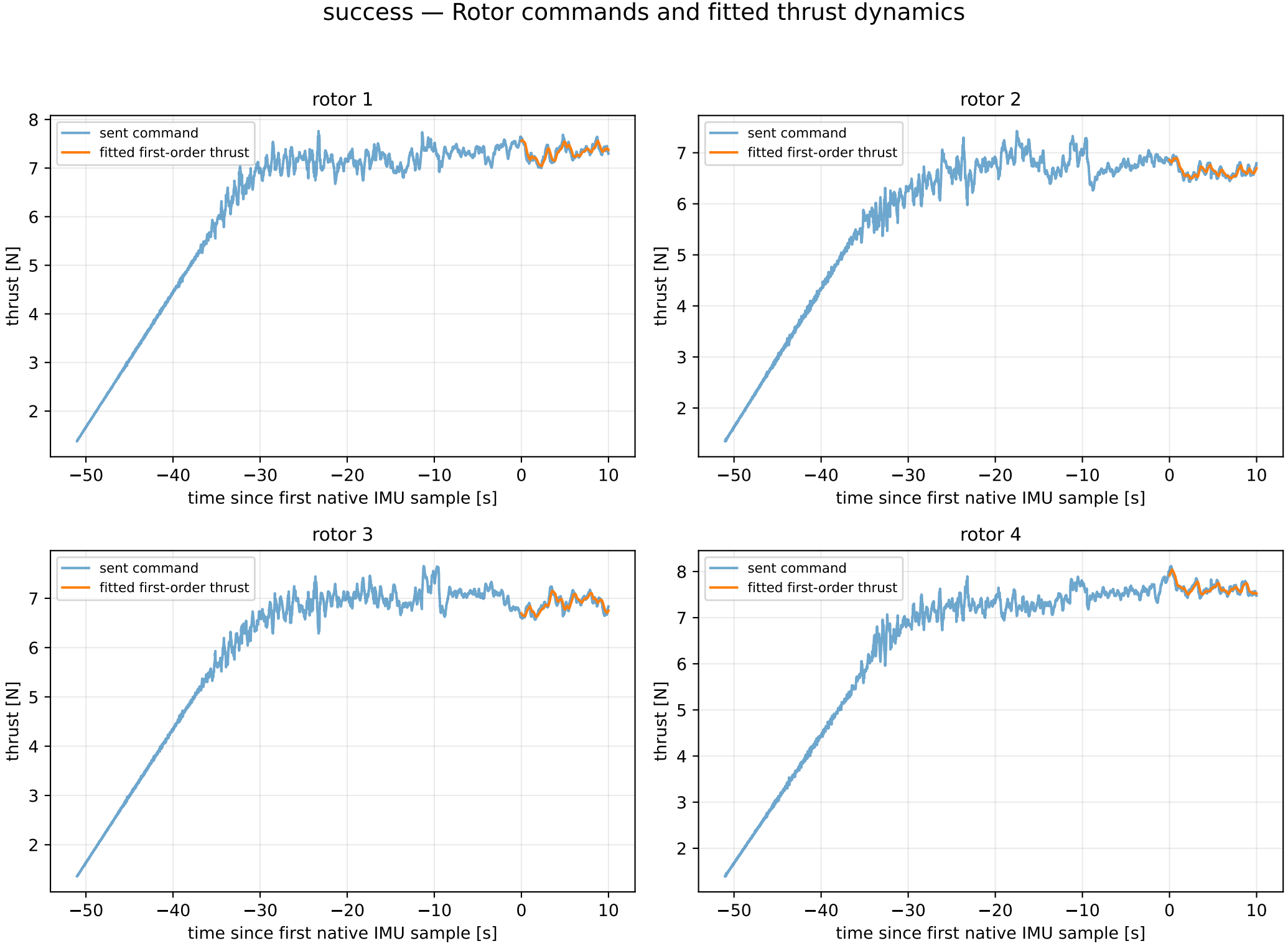
No Monte Carlo, artificial display variance, eigenvalue clipping or uncertainty-based sample selection.

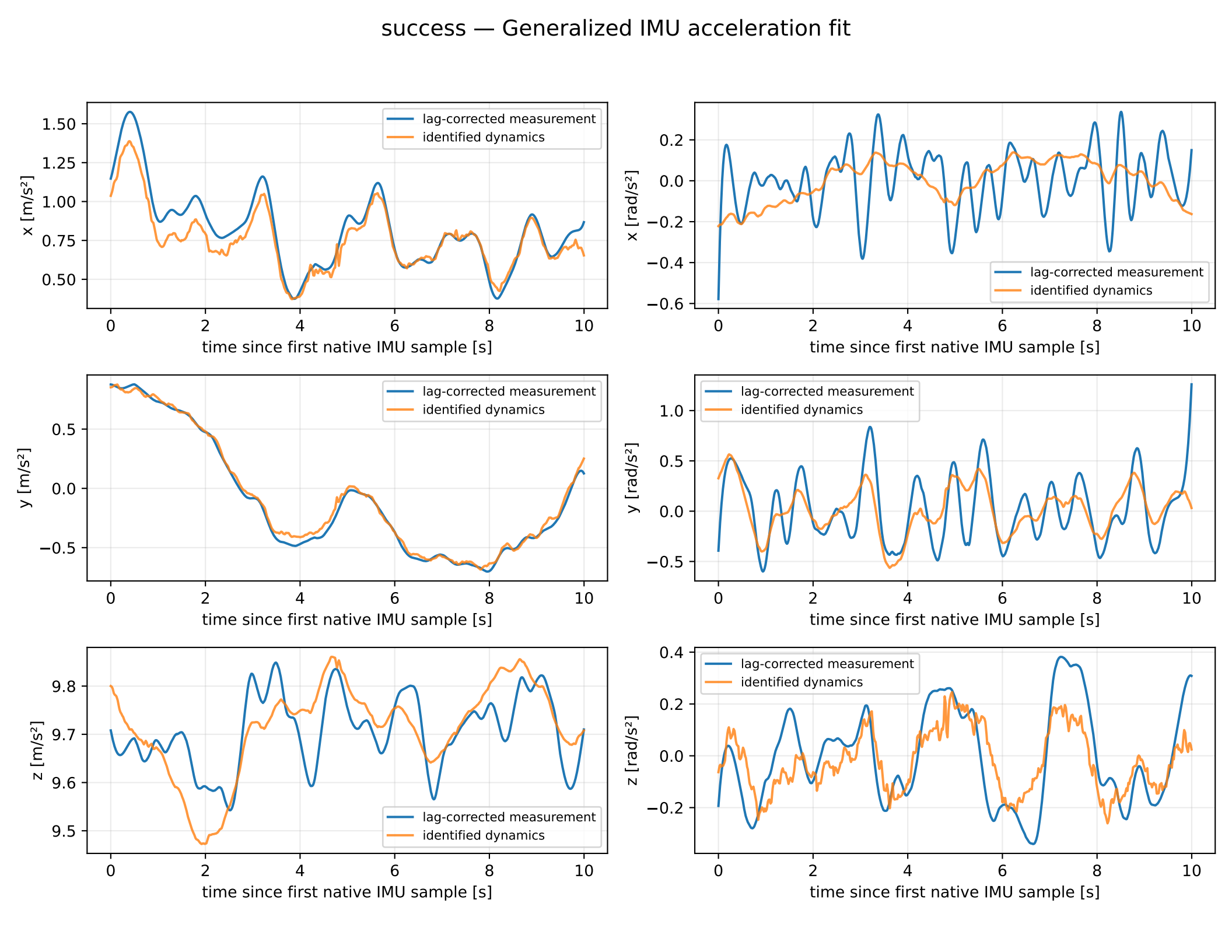
Mass / inertia / thrust scale use a common-scale gauge; chip lever arm is fixed, not calibrated.

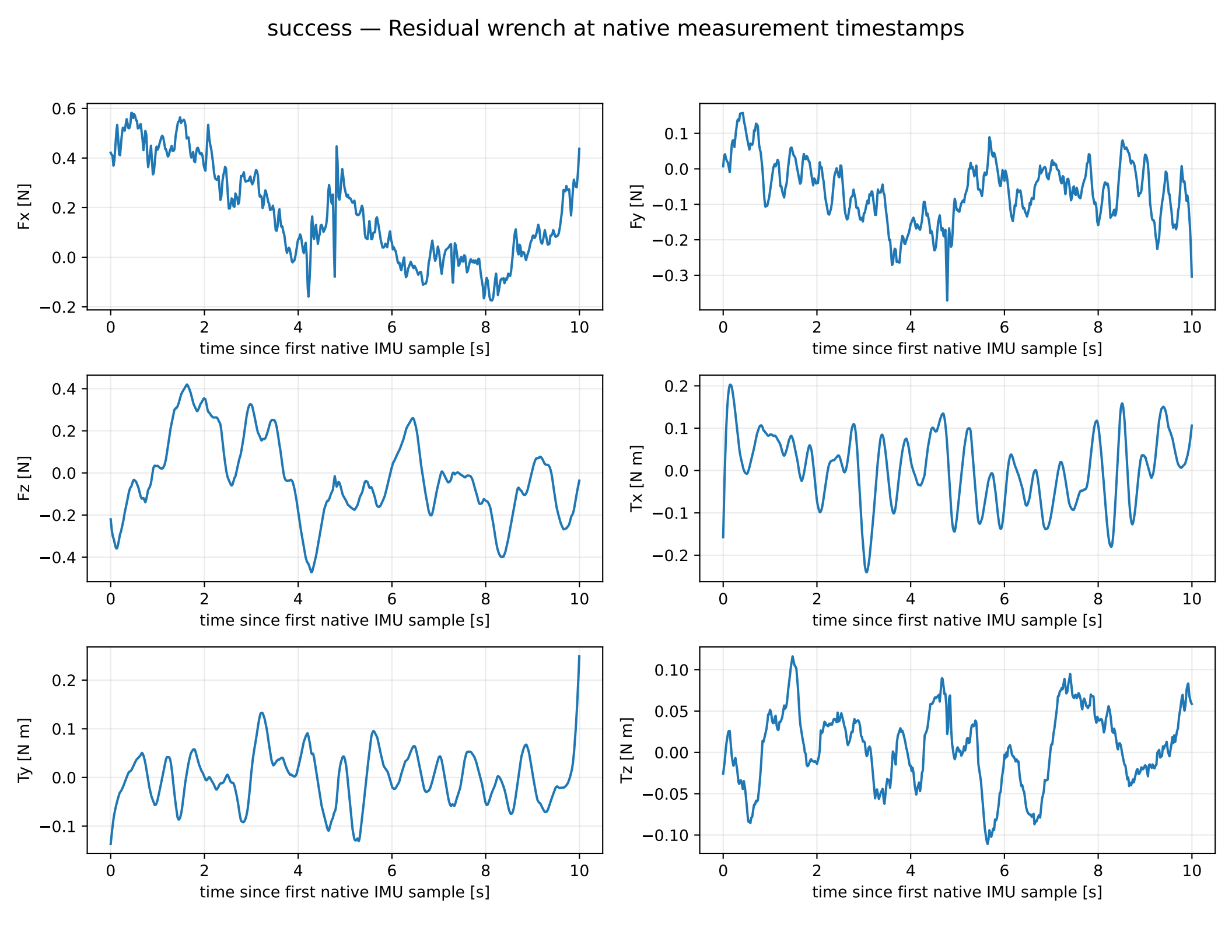
All 17 actual SI estimates and the complete joint covariance are appended at the end.

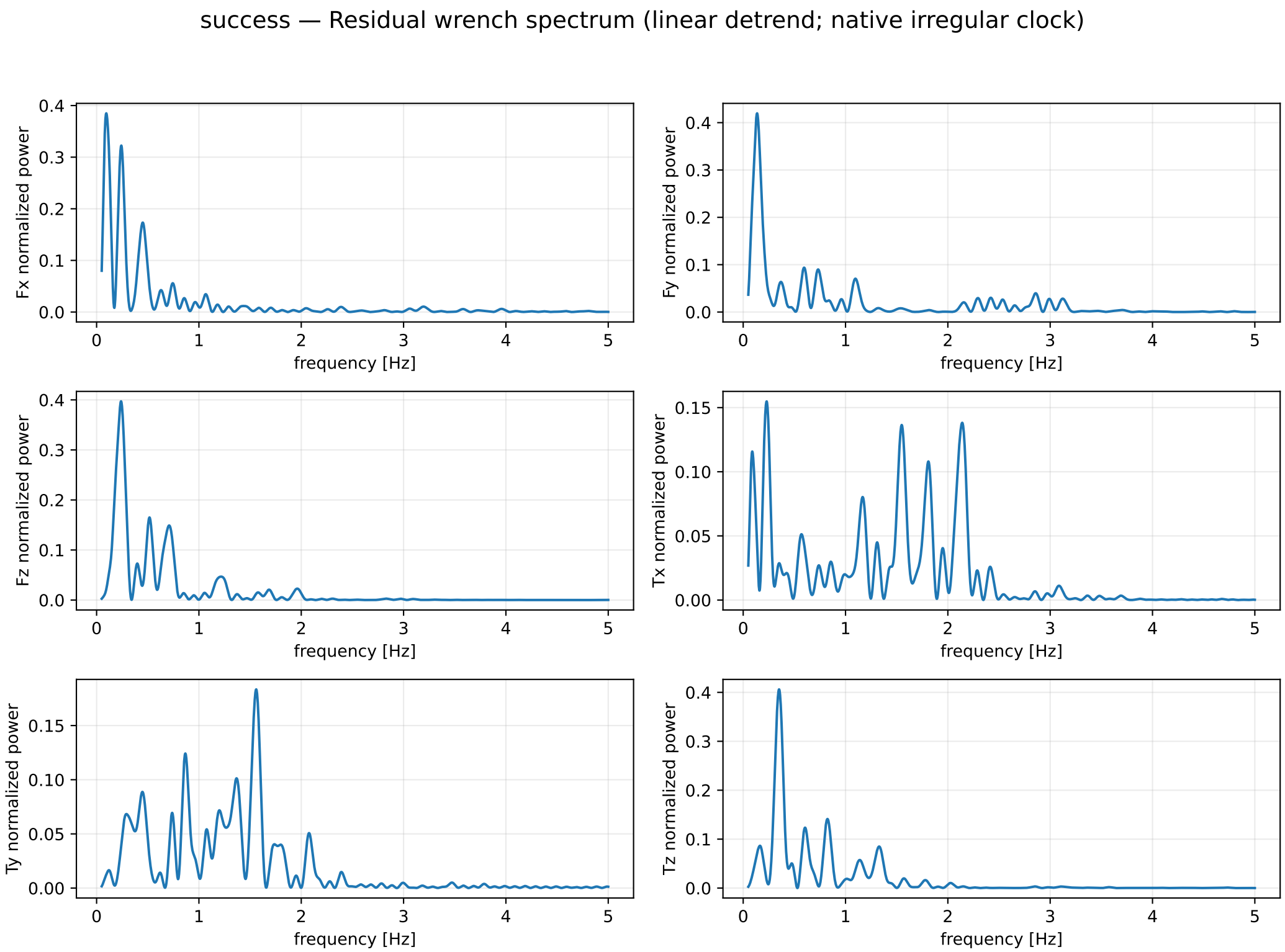
Native IMU support: 65.001922 to 74.998157 bag seconds; N = 2000.

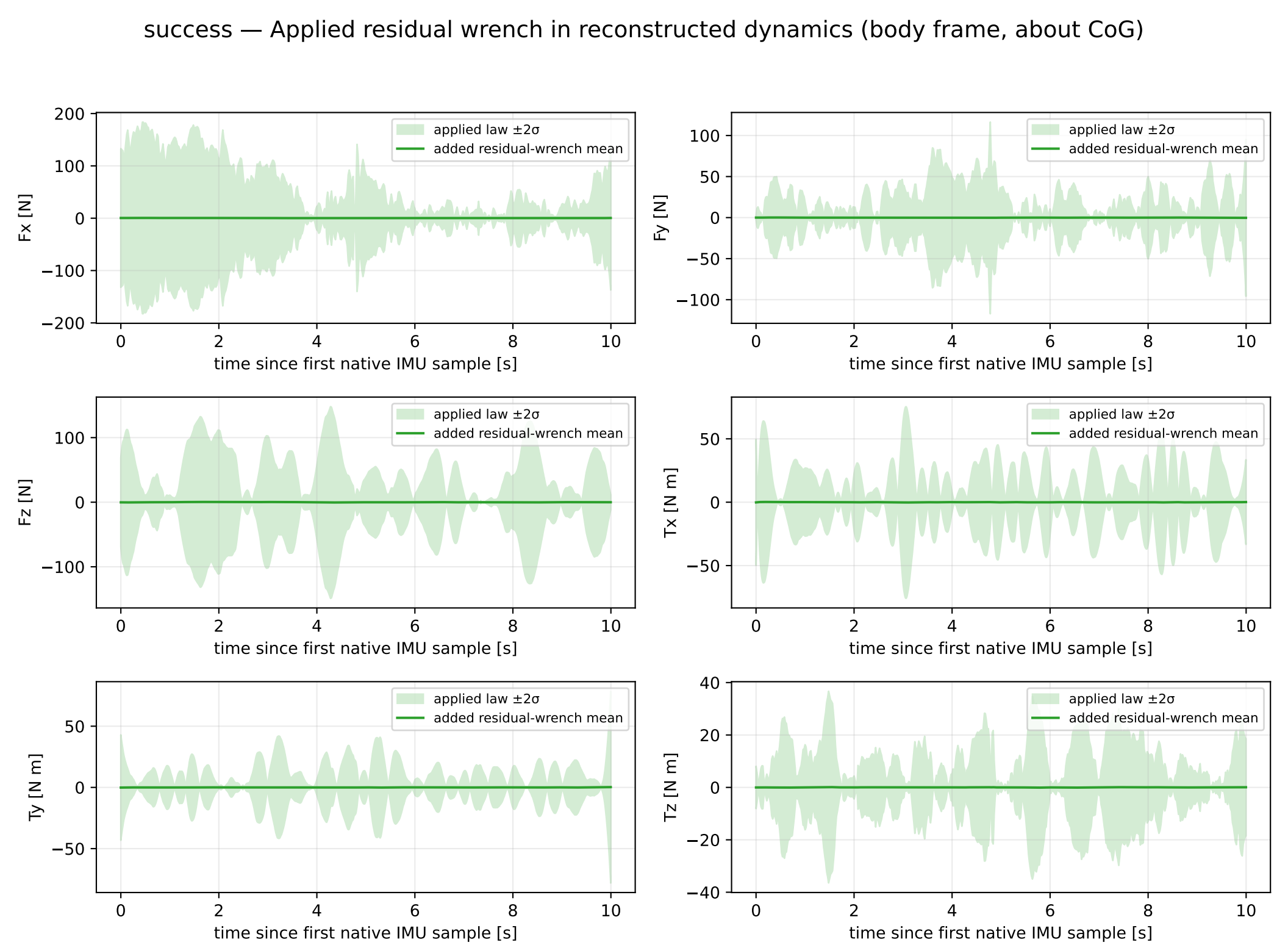
Individual PDF appendices and full-precision covariance CSV contain the same distribution.

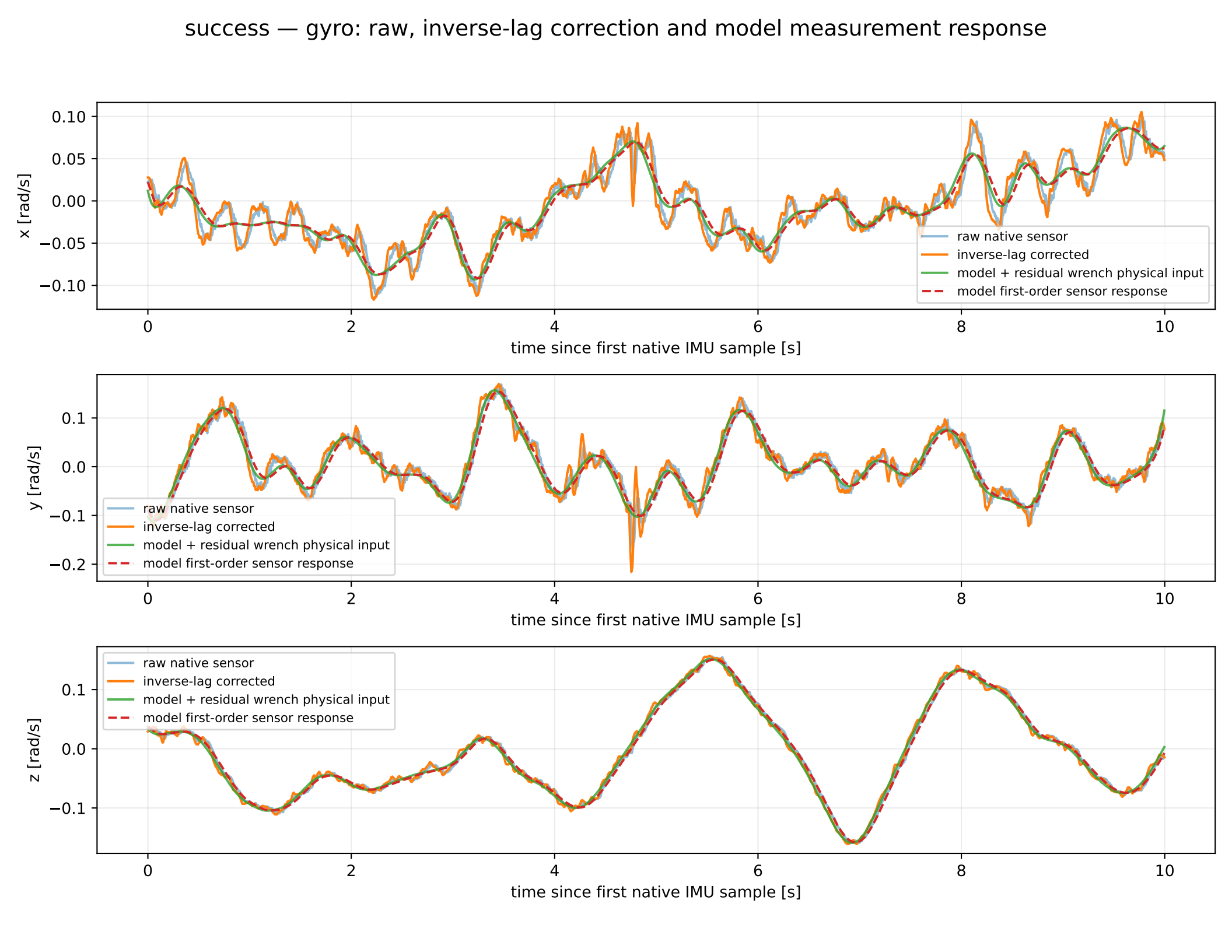

success — Rotor commands and fitted thrust dynamics
rotor 1
rotor 2
rotor 3
rotor 4
sent command
fitted first-order thrust
thrust [N]
time since first native IMU sample [s]

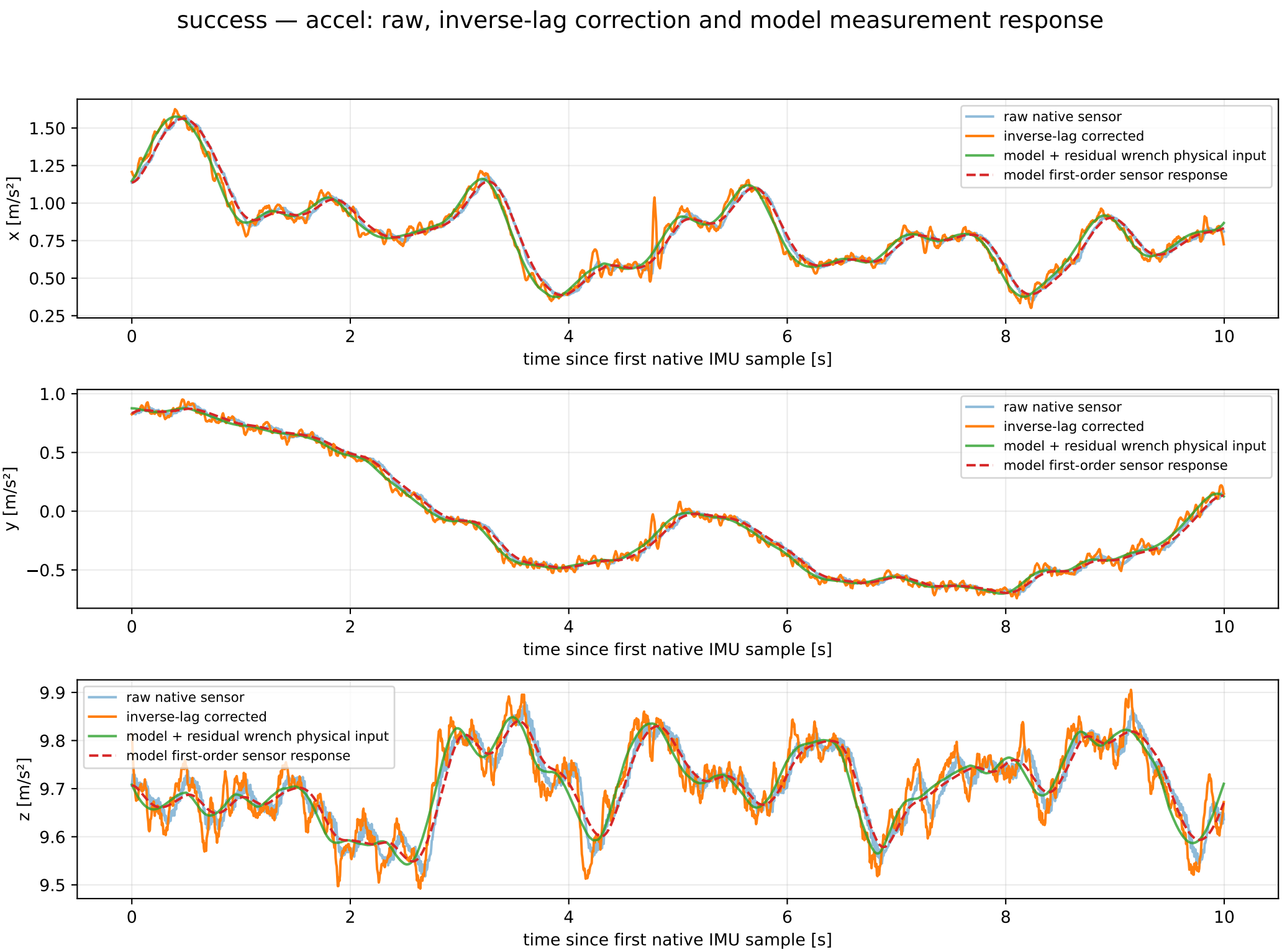

success — Generalized IMU acceleration fit
lag-corrected measurement
identified dynamics
x [m/s²]
x [rad/s²]
y [m/s²]
y [rad/s²]
z [m/s²]
z [rad/s²]
time since first native IMU sample [s]

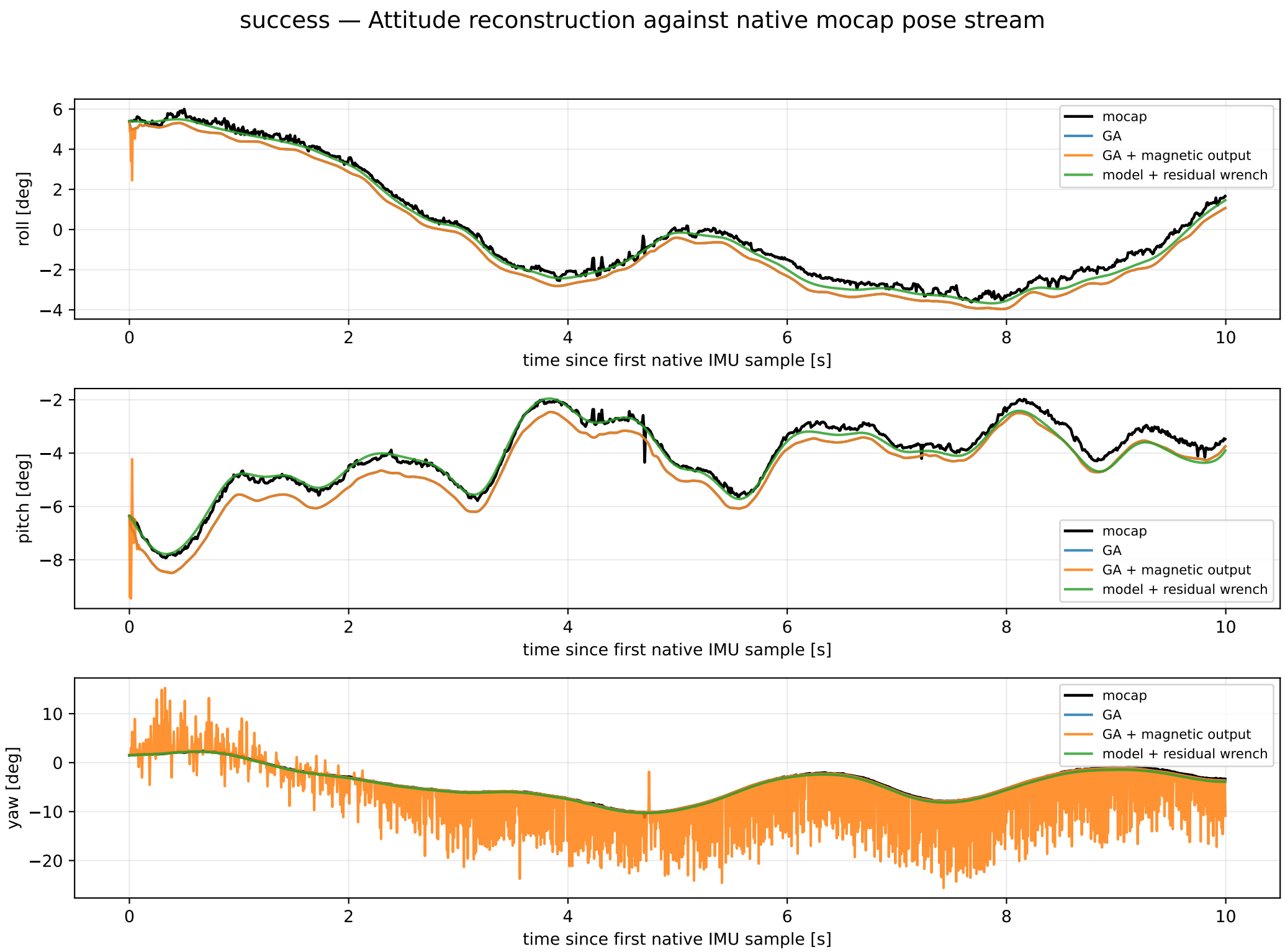

success — Residual wrench at native measurement timestamps
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time since first native IMU sample [s]

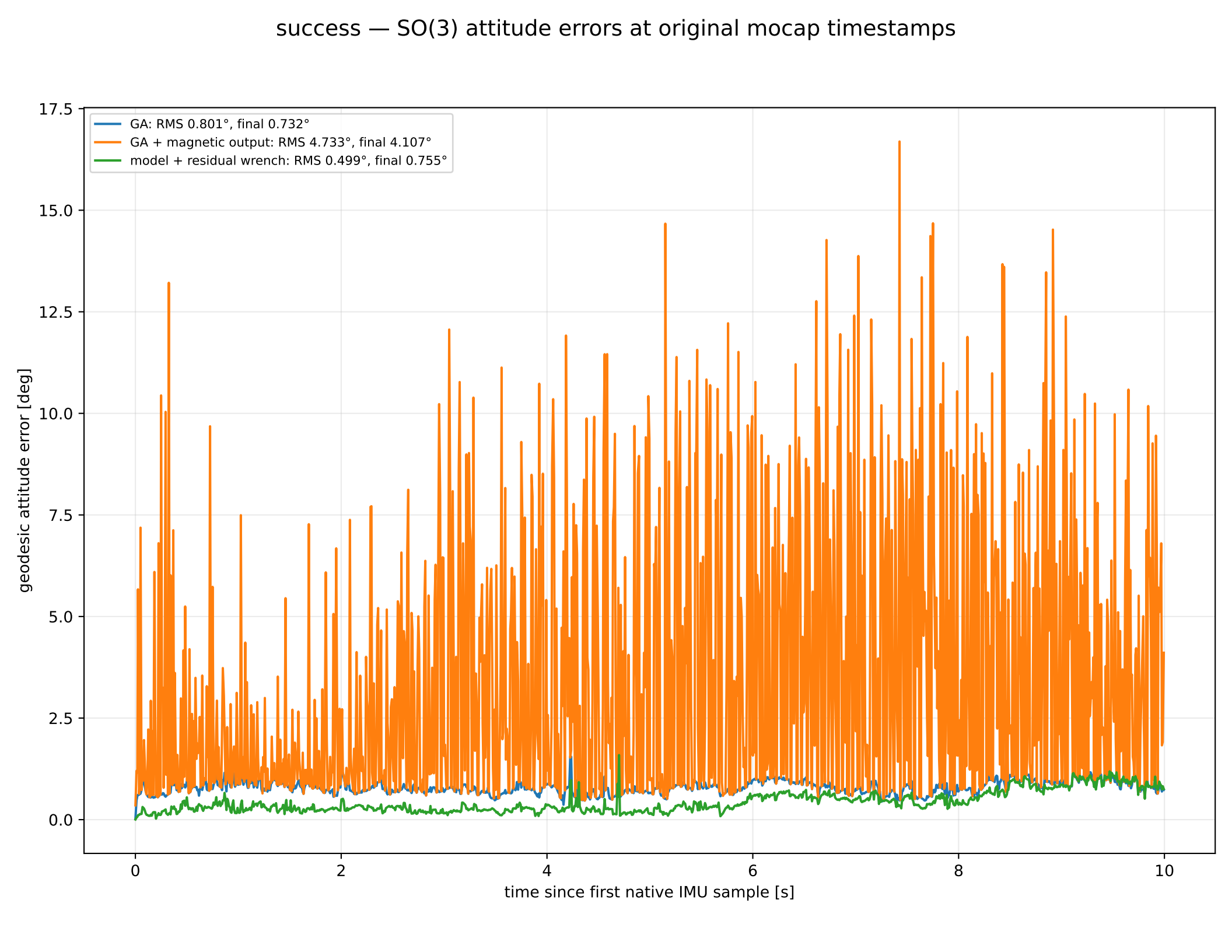

success — Residual wrench spectrum (linear detrend; native irregular clock)
Fx normalized power
Fy normalized power
Fz normalized power
Tx normalized power
Ty normalized power
Tz normalized power
frequency [Hz]

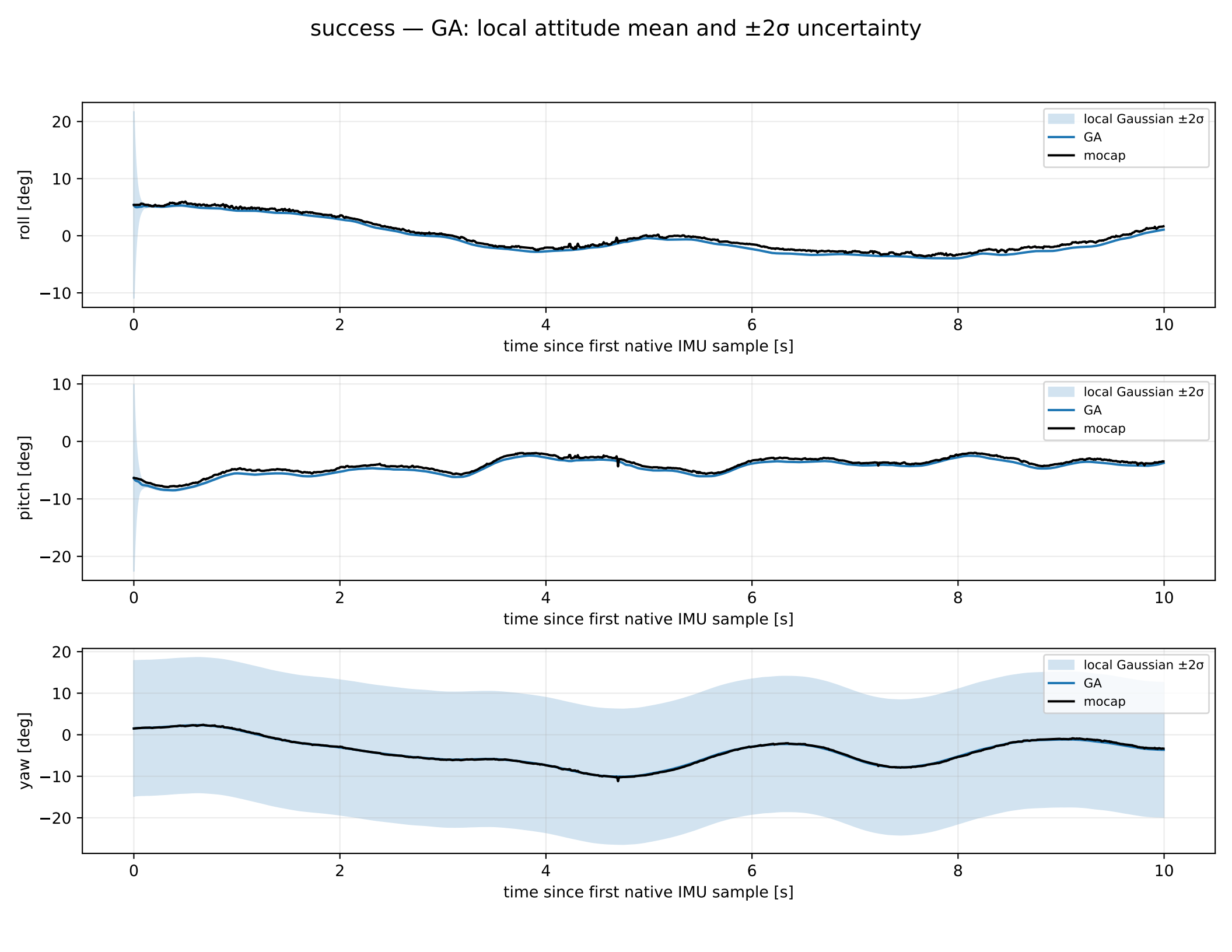

success — Applied residual wrench in reconstructed dynamics (body frame, about CoG)
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
applied law ±2σ
added residual-wrench mean
time since first native IMU sample [s]
The native-IMU residual wrench is added to the dynamics; this is not a model-only free replay.

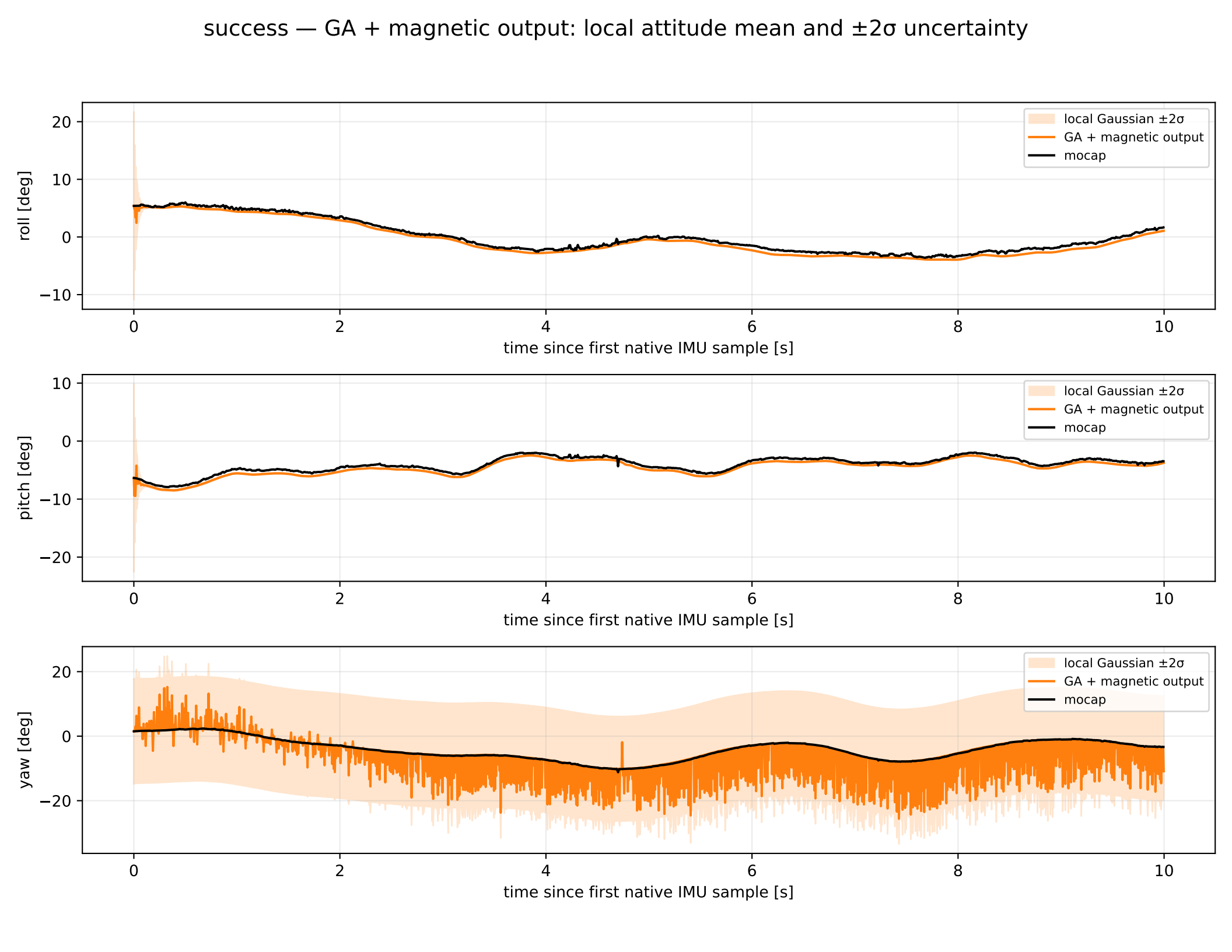

success — gyro: raw, inverse-lag correction and model measurement response
x [rad/s]
y [rad/s]
z [rad/s]
time since first native IMU sample [s]
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response

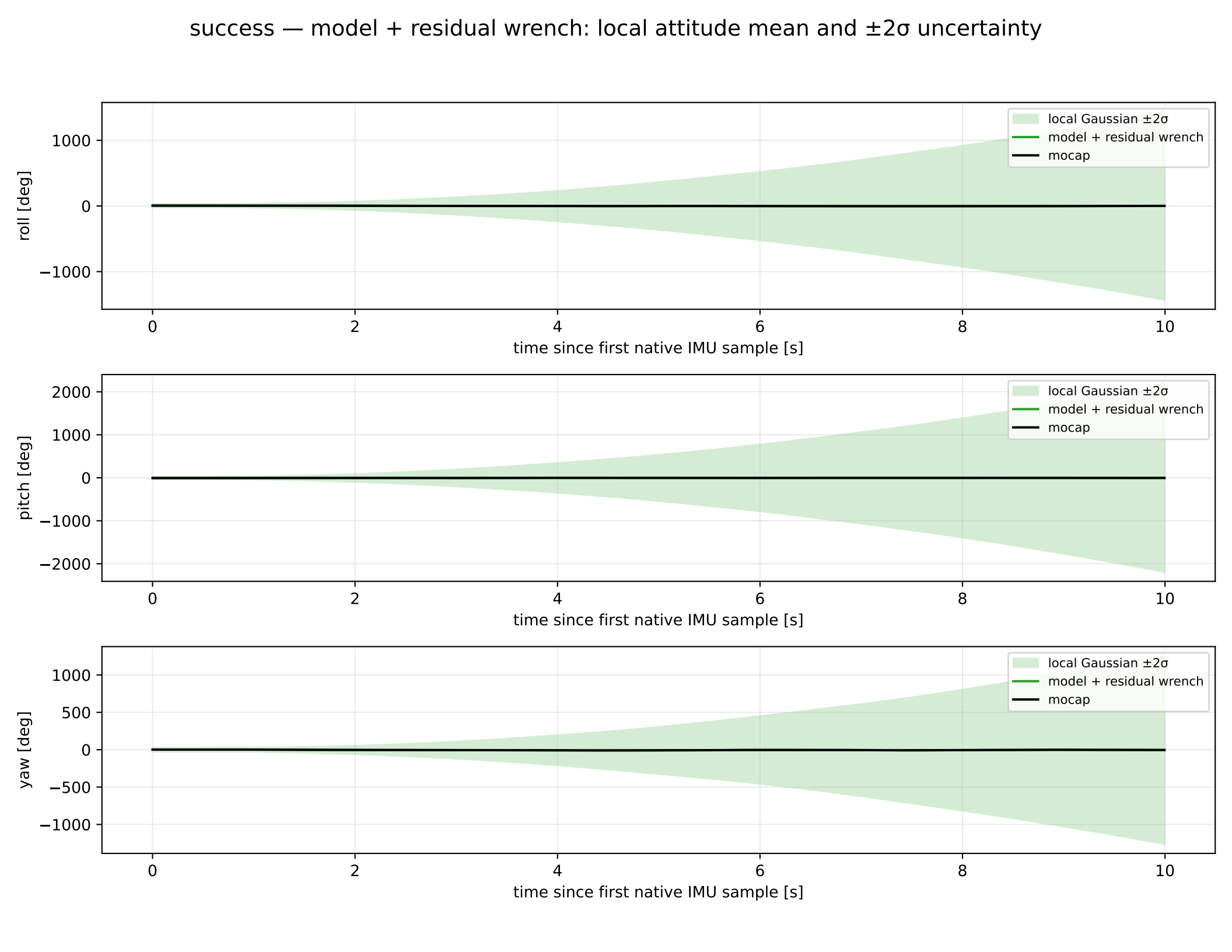

success — accel: raw, inverse-lag correction and model measurement response
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response
x [m/s²]
y [m/s²]
z [m/s²]
time since first native IMU sample [s]


success — Attitude reconstruction against native mocap pose stream
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
mocap
GA
GA + magnetic output
model + residual wrench

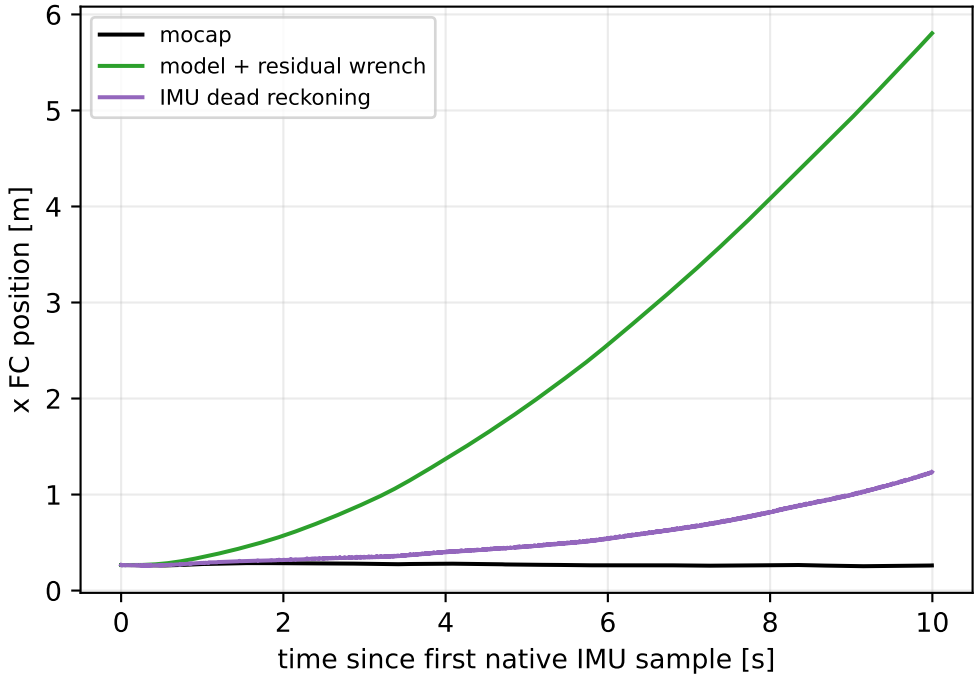

success — SO(3) attitude errors at original mocap timestamps
GA: RMS 0.801°, final 0.732°
GA + magnetic output: RMS 4.733°, final 4.107°
model + residual wrench: RMS 0.499°, final 0.755°
geodesic attitude error [deg]
time since first native IMU sample [s]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
2
4
6
8
10

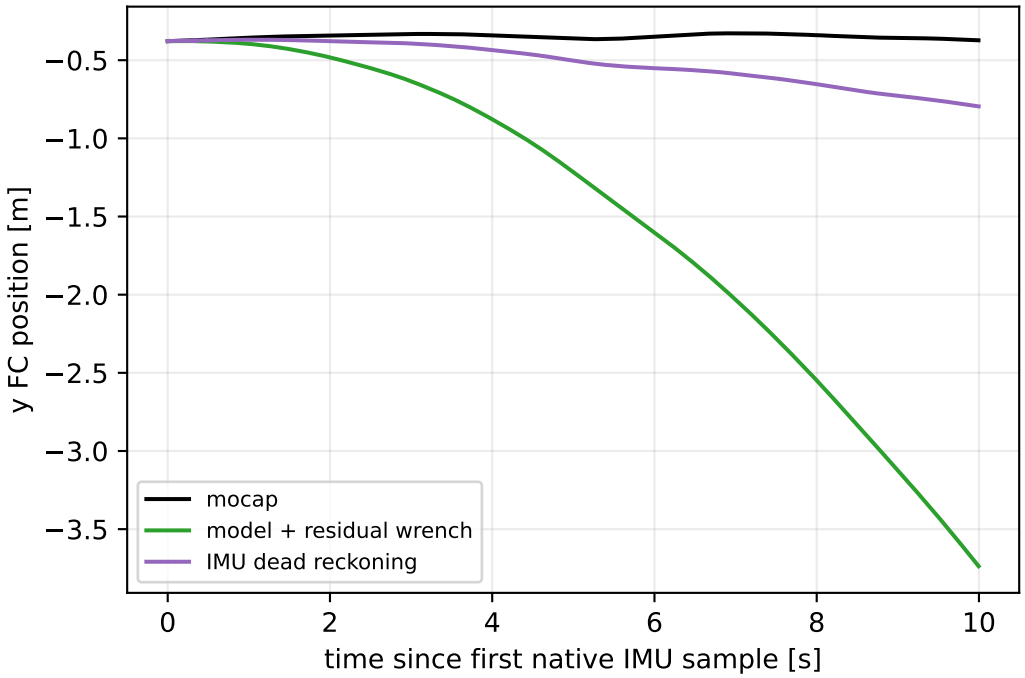

success — GA: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA
mocap

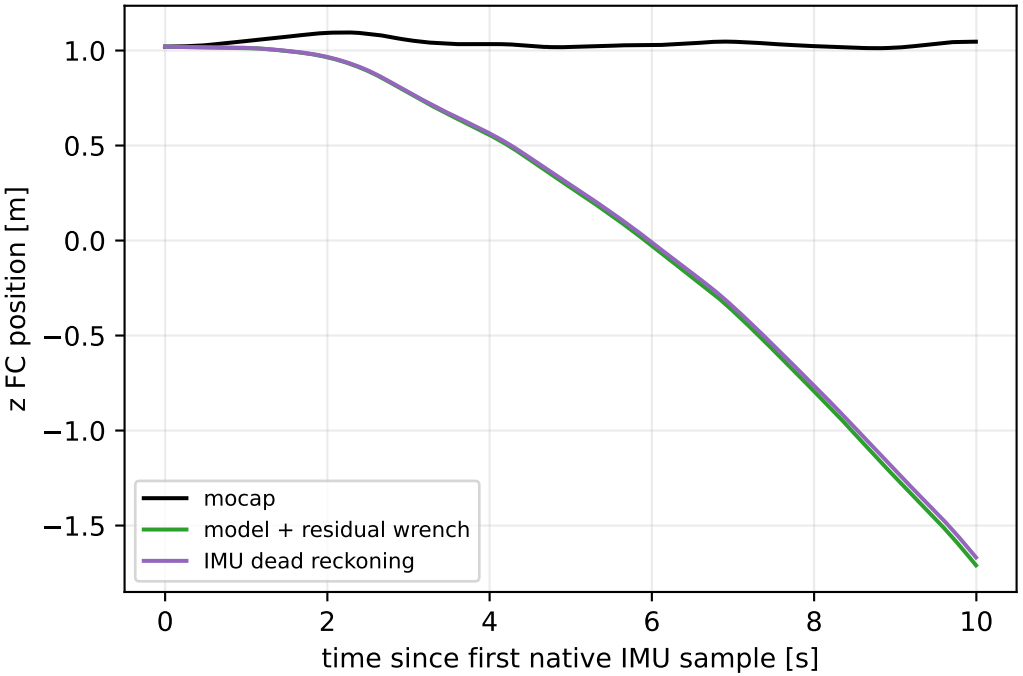

success — GA + magnetic output: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA + magnetic output
mocap

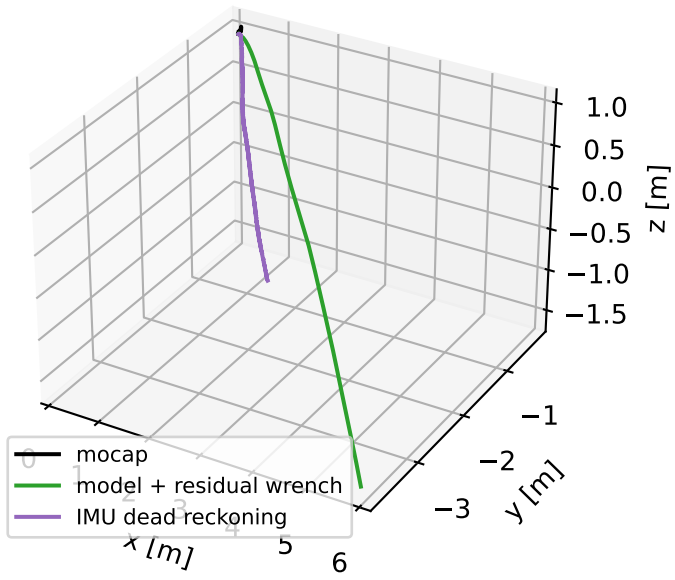

success — model + residual wrench: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
model + residual wrench
mocap

success — Position reconstruction (mean trajectories)



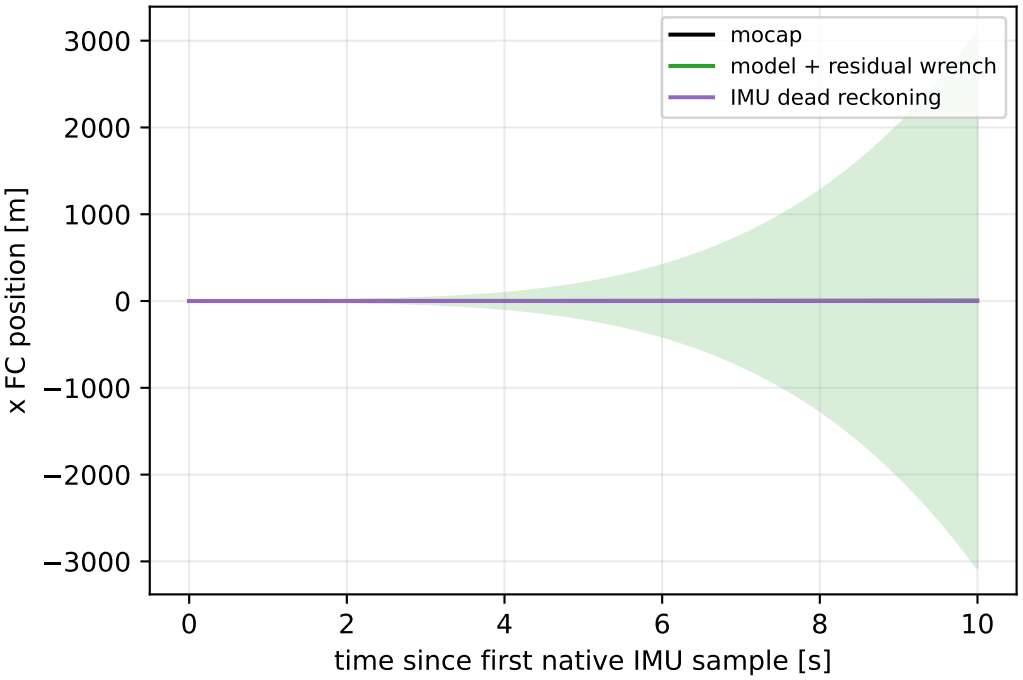


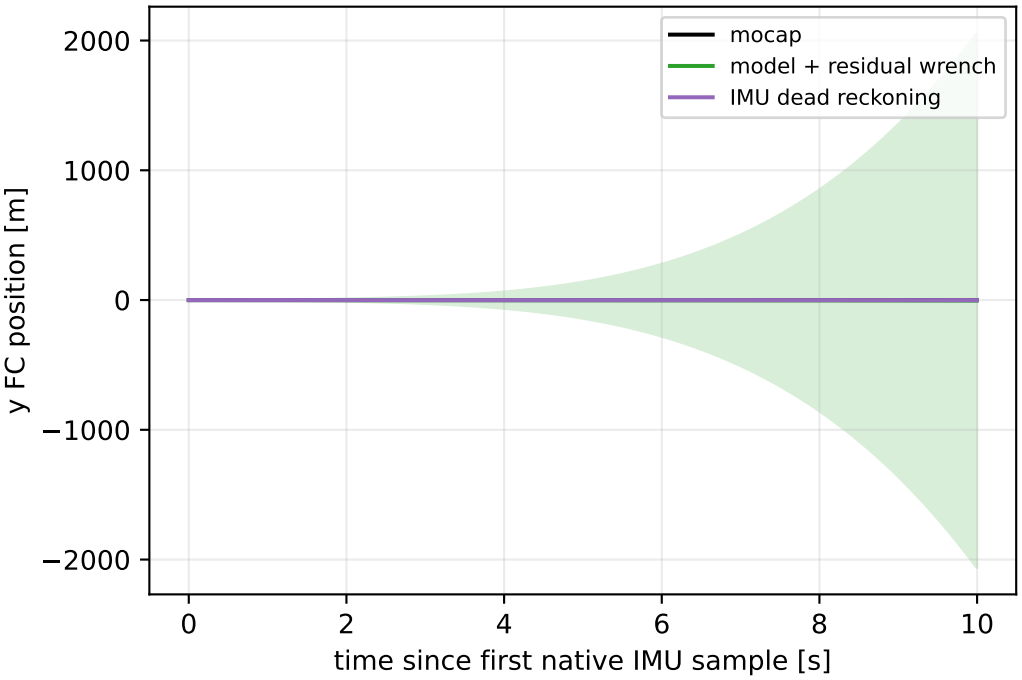


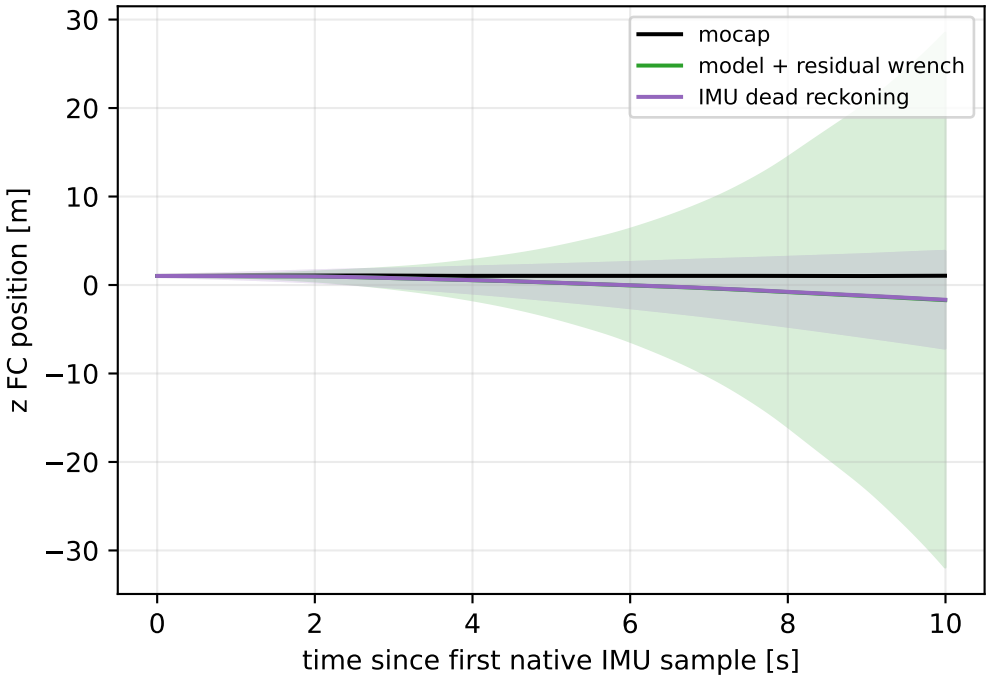


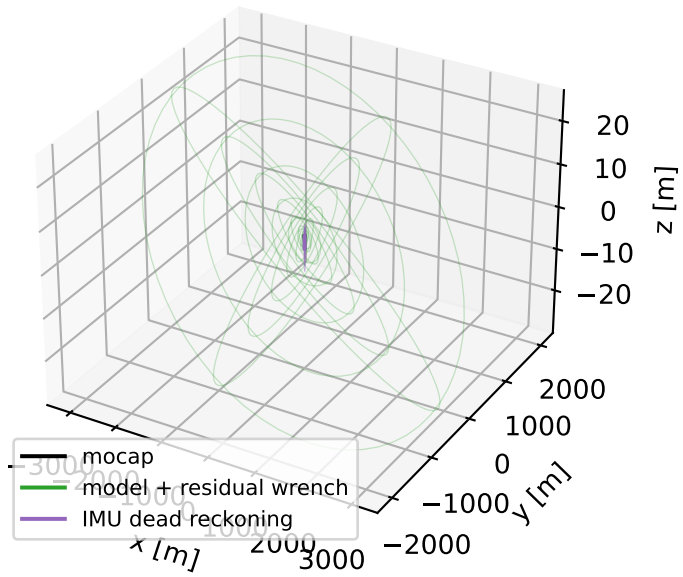
Residual-wrench model replay and IMU integration are not reset to later mocap.

success — Position reconstruction with full propagated ±2σ spread









FC origin; local Gaussian linearization. Ellipsoids: deterministic 2σ factor-SVD wireframes, not samples.

success — Actual estimated SI values and joint covariance coordinate order

#	parameter	unit	estimate	std. deviation
1	mass_kg	kg	2.858943911e+00	2.595899936e+02
2	inertia_xx_kg_m2	kg m^2	5.538736339e-01	5.032196978e+01
3	inertia_yy_kg_m2	kg m^2	2.269874855e-01	2.061466211e+01
4	inertia_zz_kg_m2	kg m^2	3.570246313e-01	3.241100546e+01
5	inertia_xy_kg_m2	kg m^2	-5.996776904e-02	5.449286735e+00
6	inertia_xz_kg_m2	kg m^2	2.349699796e-02	2.141092361e+00
7	inertia_yz_kg_m2	kg m^2	-4.059893266e-02	3.683411569e+00
8	cog_x_m	m	-1.459485598e-02	1.290555178e-03
9	cog_y_m	m	-3.770966559e-02	6.519965024e-04
10	cog_z_m	m	-7.308516943e-04	2.753033284e-03
11	force_effectiveness_1	dimensionless	1.257231281e+00	1.141591796e+02
12	force_effectiveness_2	dimensionless	1.077551658e+00	9.783795813e+01
13	force_effectiveness_3	dimensionless	8.497428069e-01	7.715238664e+01
14	force_effectiveness_4	dimensionless	7.583319663e-01	6.885846628e+01
15	thrust_time_constant_seconds	s	1.977120296e-01	1.197364424e-02
16	gyro_time_constant_seconds	s	3.093824562e-02	2.212413488e-03
17	accel_time_constant_seconds	s	6.318998861e-02	2.094005976e-03

Common-scale gauge: physical chart projected onto existing 13-D orthogonal common-scale section; tau coordinates unaffected

Covariance units are products of row/column units. All cross terms and weak modes are retained.

Full-precision values: parameter_values_si.csv; full 17×17 covariance: parameter_covariance_si.csv.

success — Full SI covariance: rows 1–9, columns 1–9

	1	2	3	4	5	6	7	8	9
1	6.738696e+04	1.306308e+04	5.351360e+03	8.413572e+03	-1.414579e+03	5.557988e+02	-9.561753e+02	-2.291420e-01	-2.213826e-02
2	1.306308e+04	2.532301e+03	1.037370e+03	1.630986e+03	-2.742186e+02	1.077425e+02	-1.853563e+02	-4.442088e-02	-4.289122e-03
3	5.351360e+03	1.037370e+03	4.249643e+02	6.681419e+02	-1.123351e+02	4.413730e+01	-7.593219e+01	-1.819796e-02	-1.757338e-03
4	8.413572e+03	1.630986e+03	6.681419e+02	1.050473e+03	-1.766167e+02	6.939409e+01	-1.193829e+02	-2.861258e-02	-2.760392e-03
5	-1.414579e+03	-2.742186e+02	-1.123351e+02	-1.766167e+02	2.969473e+01	-1.166730e+01	2.007193e+01	4.810466e-03	4.636083e-04
6	5.557988e+02	1.077425e+02	4.413730e+01	6.939409e+01	-1.166730e+01	4.584276e+00	-7.886433e+00	-1.894464e-03	-1.808476e-04
7	-9.561753e+02	-1.853563e+02	-7.593219e+01	-1.193829e+02	2.007193e+01	-7.886433e+00	1.356752e+01	3.253486e-03	3.118245e-04
8	-2.291420e-01	-4.442088e-02	-1.819796e-02	-2.861258e-02	4.810466e-03	-1.894464e-03	3.253486e-03	1.665533e-06	2.377747e-09
9	-2.213826e-02	-4.289122e-03	-1.757338e-03	-2.760392e-03	4.636083e-04	-1.808476e-04	3.118245e-04	2.377747e-09	4.250994e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

success — Full SI covariance: rows 1–9, columns 10–17

	10	11	12	13	14	15	16	17
1	-5.646762e-01	2.963458e+04	2.539775e+04	2.002799e+04	1.787497e+04	-1.463135e+00	-2.417474e-04	1.889011e-01
2	-1.094826e-01	5.744714e+03	4.923398e+03	3.882460e+03	3.465093e+03	-2.837299e-01	-4.946739e-05	3.662433e-02
3	-4.485310e-02	2.353353e+03	2.016896e+03	1.590470e+03	1.419494e+03	-1.162578e-01	-1.730520e-05	1.500415e-02
4	-7.051110e-02	3.700014e+03	3.171026e+03	2.500586e+03	2.231772e+03	-1.826853e-01	-3.242878e-05	2.358563e-02
5	1.185420e-02	-6.220856e+02	-5.331466e+02	-4.204251e+02	-3.752292e+02	3.069882e-02	4.225873e-06	-3.964057e-03
6	-4.658286e-03	2.444221e+02	2.094773e+02	1.651882e+02	1.474304e+02	-1.205187e-02	6.608024e-07	1.556730e-03
7	8.016775e-03	-4.204946e+02	-3.603769e+02	-2.841836e+02	-2.536337e+02	2.076267e-02	5.107052e-06	-2.681077e-03
8	2.577171e-06	-1.007728e-01	-8.635759e-02	-6.809979e-02	-6.078524e-02	5.341716e-06	-9.078042e-08	-8.204725e-07
9	-1.955527e-07	-9.736762e-03	-8.346371e-03	-6.577755e-03	-5.870856e-03	9.432724e-07	-3.621833e-08	6.280862e-08

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

success — Full SI covariance: rows 10–17, columns 1–9

	1	2	3	4	5	6	7	8	9
10	-5.646762e-01	-1.094826e-01	-4.485310e-02	-7.051110e-02	1.185420e-02	-4.658286e-03	8.016775e-03	2.577171e-06	-1.955527e-07
11	2.963458e+04	5.744714e+03	2.353353e+03	3.700014e+03	-6.220856e+02	2.444221e+02	-4.204946e+02	-1.007728e-01	-9.736762e-03
12	2.539775e+04	4.923398e+03	2.016896e+03	3.171026e+03	-5.331466e+02	2.094773e+02	-3.603769e+02	-8.635759e-02	-8.346371e-03
13	2.002799e+04	3.882460e+03	1.590470e+03	2.500586e+03	-4.204251e+02	1.651882e+02	-2.841836e+02	-6.809979e-02	-6.577755e-03
14	1.787497e+04	3.465093e+03	1.419494e+03	2.231772e+03	-3.752292e+02	1.474304e+02	-2.536337e+02	-6.078524e-02	-5.870856e-03
15	-1.463135e+00	-2.837299e-01	-1.162578e-01	-1.826853e-01	3.069882e-02	-1.205187e-02	2.076267e-02	5.341716e-06	9.432724e-07
16	-2.417474e-04	-4.946739e-05	-1.730520e-05	-3.242878e-05	4.225873e-06	6.608024e-07	5.107052e-06	-9.078042e-08	-3.621833e-08
17	1.889011e-01	3.662433e-02	1.500415e-02	2.358563e-02	-3.964057e-03	1.556730e-03	-2.681077e-03	-8.204725e-07	6.280862e-08

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

success — Full SI covariance: rows 10–17, columns 10–17

	10	11	12	13	14	15	16	17
10	7.579192e-06	-2.483271e-01	-2.128187e-01	-1.678269e-01	-1.497879e-01	2.229369e-05	-1.474582e-07	-2.301668e-06
11	-2.483271e-01	1.303232e+04	1.116910e+04	8.807653e+03	7.860826e+03	-6.434386e-01	-1.055746e-04	8.307276e-02
12	-2.128187e-01	1.116910e+04	9.572266e+03	7.548432e+03	6.736972e+03	-5.514531e-01	-9.153034e-05	7.119410e-02
13	-1.678269e-01	8.807653e+03	7.548432e+03	5.952491e+03	5.312595e+03	-4.348544e-01	-7.222006e-05	5.614351e-02
14	-1.497879e-01	7.860826e+03	6.736972e+03	5.312595e+03	4.741488e+03	-3.881062e-01	-6.415081e-05	5.010837e-02
15	2.229369e-05	-6.434386e-01	-5.514531e-01	-4.348544e-01	-3.881062e-01	1.433682e-04	-1.113262e-05	-7.428792e-06
16	-1.474582e-07	-1.055746e-04	-9.153034e-05	-7.222006e-05	-6.415081e-05	-1.113262e-05	4.894773e-06	6.262505e-07
17	-2.301668e-06	8.307276e-02	7.119410e-02	5.614351e-02	5.010837e-02	-7.428792e-06	6.262505e-07	4.384861e-06

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.